AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Proteins and DNA

Proteins do most of the work in a cell, and DNA holds the instructions for building them. This topic follows the story from amino acids, through the folding of a polypeptide into a working protein, to the double helix of DNA and the way it copies itself before each cell division.

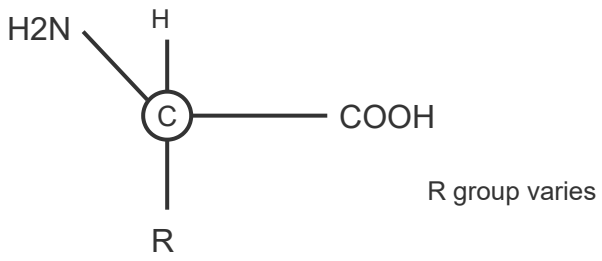
SPEC 2.6 Amino acids and protein structure

WHAT THIS SPEC POINT ASKS

(i) Know the basic structure of an amino acid. (ii) Understand how polypeptides form by condensation, making peptide bonds. (iii) Understand how primary structure determines secondary, tertiary (3-D) structure and properties, including globular vs fibrous proteins and the bonds involved. (iv) Know the structure of a globular protein (haemoglobin) and a fibrous protein (collagen) and how it relates to function.

Amino acids and peptide bonds

- Every amino acid has a central carbon attached to: an **amine group (–NH_2)**, a **carboxyl group (–COOH)**, a **hydrogen**, and an **R group** (side chain). There are **20** amino acids, differing only in the R group. (Specific R groups don't need learning.)
- **Condensation**: the –OH of one amino acid's carboxyl group and an –H from the next one's amine group are removed as **water**, forming a **peptide bond** (–CO–NH–). Two make a **dipeptide**; many make a **polypeptide**.
- **Hydrolysis** (by protease enzymes) adds water and breaks the peptide bond.
- **Calculation**: n amino acids are joined by n − 1 peptide bonds, so the mass of a polypeptide = total mass of the amino acids − 18 × (n − 1), one water lost per bond.



Four levels of structure

Level	What it is	Bonds holding it
Primary	the **sequence** of amino acids (set by the gene)	peptide bonds
Secondary	the chain folds into an **α-helix** or a **β-pleated sheet**	**hydrogen bonds** between C=O and N–H groups of the backbone
Tertiary	the whole chain folds into a specific **3-D shape**	bonds between **R groups**: hydrogen bonds, **ionic bonds**, **disulfide bridges** (between cysteines; strong, covalent), hydrophobic interactions
Quaternary	two or more polypeptide chains together (sometimes with a prosthetic group, e.g. haem)	same as tertiary

Why primary structure matters: the sequence decides which R groups are where, so it decides where the bonds form, so it decides the **3-D shape**, which decides the **function** (e.g. the active site of an enzyme). Change one amino acid (a mutation, 2.14) and the shape can change; e.g. in **sickle cell anaemia** one polar amino acid in haemoglobin (**glutamic acid**) is replaced by a non-polar one (**valine**), and the haemoglobin clumps.

Globular vs fibrous

	Globular (e.g. haemoglobin, enzymes, insulin)	**Fibrous** (e.g. collagen, keratin)
Shape	compact, spherical	long, thin strands
R groups	**hydrophobic inside, hydrophilic outside**	repetitive sequence
Solubility	**soluble** in water	**insoluble**
Role	metabolic: transport, enzymes, hormones	**structural**: strength

Haemoglobin (globular)

- **Quaternary** structure: **four** polypeptide chains (2 α, 2 β), each with a **haem** prosthetic group containing **Fe^{2+}** that binds O_2.
- Hydrophilic R groups on the outside → **soluble** → carried in solution inside red blood cells.
- Its shape changes slightly as O_2 binds (cooperative binding, 1.9).

Collagen (fibrous)

- Three polypeptide chains wound into a **triple helix** (a rope-like structure).
- Every **third amino acid is glycine** (the smallest), so the chains can pack **tightly**. The sequence repeats and is rich in **proline** and **hydroxyproline**.
- **Hydrogen bonds** between the chains, and **covalent cross-links** between neighbouring molecules, form **fibrils**; the ends of molecules are **staggered** so there are no weak points. Fibrils bundle into **fibres**.
- Result: very high **tensile strength** and insoluble → found in tendons, skin, bone and **artery walls**.

CORE PRACTICAL

Recommended practical: biuret test for protein

Add biuret reagent (copper(II) sulfate in alkaline solution) to the sample. **Purple/lilac** = protein (it reacts with peptide bonds); stays **blue** = no protein. Using colour standards of known protein concentration makes it semi-quantitative, as in CP1.

MARK-SCHEME WORDING

Haemoglobin: "**four haem groups**" (iron only credited if linked to haem) · "globular, **polar/hydrophilic R groups outside**" · "**soluble in the cytoplasm** of red blood cells" (**"soluble in blood" is not accepted**) · "cooperative binding between subunits"

"sequence of amino acids" (primary) · "hydrogen bonds **between amino acids**" (secondary; **not** "between R groups") · "bonds **between R groups**: hydrogen, ionic, disulfide" (tertiary; naming a wrong bond loses the mark) · "more than one polypeptide" (quaternary) · "hydrophilic R groups on the outside so soluble"

Bond name: "**peptide**". **Not accepted**: "dipeptide bond". (Avoid "amide": accepted once, rejected later.)

Drawing an amino acid: central C **with its H**, amine, carboxyl and R group. Check every C has 4 bonds, N 3, O 2, H 1.

Collagen: "fibrous", "**three polypeptide chains** / triple helix", "hydrogen bonds **between the chains**", "every third amino acid is glycine" → tight packing, "cross-links" → tensile strength.

Heating: "**hydrogen bonds break**", R groups vibrate more; "**peptide bonds are not affected**", so the primary structure is unchanged. A thermostable protein has "more / stronger bonds", so "more energy is needed to break them".

Often asked as: draw an amino acid or a peptide bond; describe collagen's structure; explain why a changed amino acid stops a protein working; explain thermostability.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Relative bond strengths**: disulfide (covalent) > ionic > hydrogen. More disulfide or ionic bonds → more energy needed to break the tertiary structure → more **thermostable** (WBI11 Jun 2025).
- **Sickle cell haemoglobin (HbS)**: valine's non-polar R group turns away from water and forms **hydrophobic interactions** with other haemoglobin molecules, so they stick together in the cytoplasm of red blood cells (WBI11 Jan 2022).
- **Collagen mutations**: replacing a glycine with an amino acid with a larger R group stops tight packing and weakens the triple helix (osteogenesis imperfecta) (WBI11 Jan 2025).
- **Proinsulin → insulin**: a protease hydrolyses peptide bonds to cut out a section of the chain (WBI11 Oct 2022).
- Proteins contain **C, H, O, N** and often **S** (WBI11 Oct 2025).

COMMON MISTAKE

"**Dipeptide bond**" is the commonest error. It's a **peptide** bond.

Leaving the H off the central carbon when drawing an amino acid.

Not saying where hydrogen bonds are: **between the chains** (collagen) or **between amino acids** (secondary structure).

Writing "iron" for haemoglobin: say "**iron ion (Fe^{2+})** in a haem group".

SPEC 2.7 Enzymes: how they work

WHAT THIS SPEC POINT ASKS

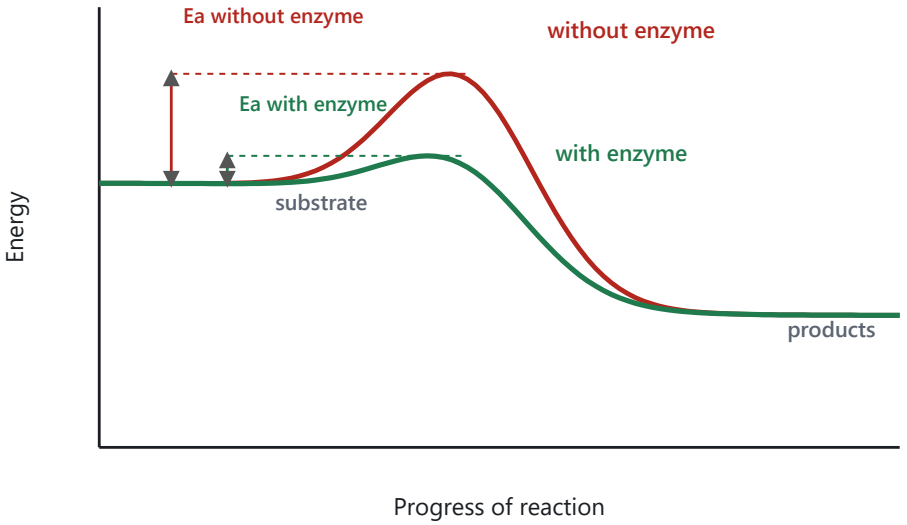
(i) Understand how enzymes work, and their specificity, in terms of their 3-D structure. (ii) Understand that enzymes are biological catalysts that lower activation energy. (iii) Know that intracellular enzymes work inside cells and extracellular enzymes outside them.

Mechanism and specificity

- Enzymes are **globular proteins**. Their tertiary structure creates an **active site** with a specific shape.
- Only a substrate with a **complementary** shape can fit and bind, forming an **enzyme–substrate complex**. That's why each enzyme is **specific** to one substrate (or one type of reaction).
- **Induced fit** model: the active site **changes shape slightly** as the substrate binds, moulding around it. This puts strain on the substrate's bonds (or brings reacting groups close together), making the reaction easier. Products are released and the enzyme is **unchanged**, ready to be reused.
- (The older **lock-and-key** model treated the active site as rigid. Induced fit replaced it because it explains why binding changes the enzyme's shape.)

Catalysts that lower activation energy

- **Activation energy** is the energy needed to start a reaction.
- Enzymes **lower the activation energy**, so reactions happen fast at body temperature, which would otherwise need far higher temperatures.
- They don't change the overall energy difference between substrate and product, and they aren't used up.



The enzyme lowers the activation energy (Ea). Starting and finishing energies are unchanged.

Where enzymes work

Intracellular (inside cells)	**Extracellular** (secreted, work outside cells)
e.g. **catalase** (breaks down toxic hydrogen peroxide), enzymes of respiration, DNA polymerase, RNA polymerase	e.g. **amylase** (starch → maltose), **trypsin** and **pepsin** (proteins), **lipase** (lipids) in digestion
made and used in the same cell	made in the cell, released by **exocytosis**

MARK-SCHEME WORDING

Biological catalyst: define **both** words: "**protein** found in living organisms" + "**speeds up** a reaction **without being used up** / lowers the activation energy".

Mechanism: substrate "**binds to the active site**" → "**enzyme–substrate complex**" → activation energy lowered → bond broken by **hydrolysis**; name the bond (peptide, glycosidic). "Complementary" on its own isn't enough.

Specificity: the active site's shape depends on its **R groups**; the other substrate has a "**different shape / is not complementary**". Deal with **both** substrates.

Activation-energy diagrams: Ea is lower with the enzyme; the overall energy change is the **same**.

Often asked as: activation-energy diagram MCQs; why an enzyme acts on one substrate but not a similar one; enzyme action in a new context (lactase, proteases); inhibitors.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Competitive inhibitor** (WBI11 Oct 2025): shape **similar to the substrate**, so it **binds to the active site** → "**fewer enzyme–substrate complexes** formed per unit time". Adding more substrate outcompetes it, so the maximum rate can still be reached.
- **Non-competitive inhibitor** (WBI11 Oct 2025): binds **elsewhere on the enzyme**, at an **allosteric site** → "the **active site changes shape**" → fewer active sites the substrate can bind to. More substrate doesn't help: the maximum rate stays lower.
- **Warfarin** is a competitive inhibitor: its structure is similar to vitamin K, so it blocks the active site of the enzyme that recycles vitamin K for making clotting factors. Extra vitamin K competes with it. "Non-competitive" is not accepted (WBI11 Oct 2020).
- **Metabolic pathways**: the product of one enzyme is the substrate of the next (WBI11 Oct 2019).
- **Active-site size**: a larger active site can bind more substrate per unit time but may hold it less tightly (WBI11 Jun 2025).

COMMON MISTAKE

"The substrate and active site have the **same** shape." They have **complementary** shapes. And enzymes are **denatured**, not "killed".

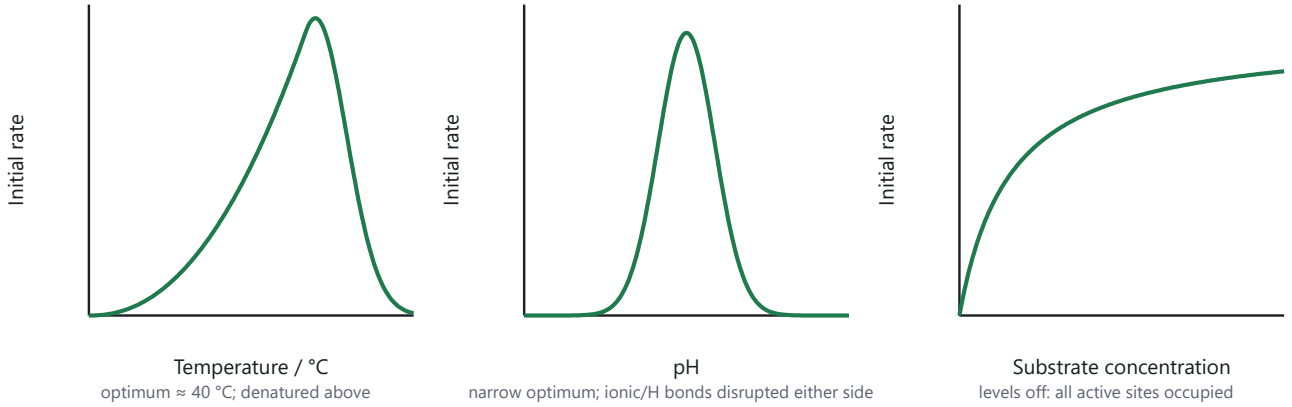
Putting the active site on the **substrate**. Mixing up **lactose** (the substrate) and **lactase** (the enzyme).

Saying a protease breaks hydrogen or disulfide bonds: it hydrolyses **peptide** bonds.

SPEC 2.8 Core Practical 4: rate of enzyme-controlled reactions

WHAT THIS SPEC POINT ASKS

Investigate how temperature, pH, enzyme concentration and substrate concentration affect the initial rate of enzyme-catalysed reactions.



Typical shapes of the curves for the four factors (enzyme concentration gives a straight line while substrate is in excess).

How each factor works

- **Temperature**: as temperature rises, molecules have more **kinetic energy** → more frequent **successful collisions** between enzyme and substrate → more enzyme–substrate complexes → faster rate. Above the **optimum**, bonds holding the tertiary structure (especially hydrogen bonds) break → the active site changes shape → substrate no longer fits → enzyme **denatured** → rate falls fast. Denaturation is permanent.
- **pH**: the **H^+ / OH^-** concentration changes the **ionisation (charge) of R groups**, which affects the **ionic and hydrogen bonds** that hold the tertiary structure (and charges in the active site). Away from the optimum pH the active site changes shape; at extremes the enzyme denatures. Different enzymes have different optima (pepsin ≈ pH 2, amylase ≈ pH 7).
- **Enzyme concentration**: more enzyme → more active sites → rate rises **in proportion**, as long as substrate is in excess.
- **Substrate concentration**: rate rises at first (more substrate molecules → greater chance of **collisions with active sites**), then **levels off** when all active sites are **occupied** (saturated); enzyme concentration is now the limiting factor.

Measuring initial rate

- Rate is fastest at the start, because substrate is used up as the reaction goes on. So compare **initial rates**.
- Plot product formed (or substrate left) against time, draw a **tangent** to the curve at **t = 0**, and find its gradient.

WORKED EXAMPLE

Worked example: initial rate from a tangent

O_2 produced by catalase is plotted against time. The tangent at t = 0 passes through (0 s, 0 cm^3) and (20 s, 9.0 cm^3).

Initial rate = change in y ÷ change in x = 9.0 ÷ 20 = **0.45 cm^3 s^{-1}**.

Rate can also be found as **1 ÷ time** to reach an end point (e.g. time for iodine colour to disappear), units s^{-1}.

CORE PRACTICAL

Core practical 4: two standard set-ups

Catalase + hydrogen peroxide (e.g. enzyme concentration): mix a **buffer**, H_2O_2 and catalase (or potato/yeast extract) in a conical flask connected to an upside-down measuring cylinder (or gas syringe) to collect O_2. Read the O_2 volume every 10 s for 60 s. Repeat for each enzyme concentration, 3 times each.

Amylase + starch (e.g. temperature or pH): mix starch and amylase (pre-incubated at the test temperature/pH in a water bath with a **buffer**). Every 10–30 s, put a drop onto **iodine** on a spotting tile; record the time when it stays **orange-brown** (all starch digested). Or follow the fall in blue-black colour with a colorimeter.

Control variables: the other three factors (temperature with a **water bath**, pH with a **buffer**, and the volumes and concentrations), and the source and age of the enzyme. Leave solutions to reach temperature before mixing (**equilibrate**).

MARK-SCHEME WORDING

Rising part of a rate–substrate curve: "more substrate molecules" → "greater chance of **collisions with the active site**" → more enzyme–substrate complexes. A bare "more collisions" is ignored.

Plateau: "**all active sites are filled** / enzyme is saturated". "Substrate is used up" is ignored.

High temperature: kinetic energy → "R groups vibrate" → bonds break → "active site distorted" → "enzyme–substrate complexes can't form". Write "**denatured**", not "starts to denature".

pH: "changes the **ionisation of R groups**" → bonds break → active site changes shape.

Rate = change in product (or substrate) **per unit time**, with units, e.g. $cm^3\ s^{-1}$ or $mmol\ dm^{-3}\ min^{-1}$. No time unit (or "a.u." alone) loses the mark.

Sketching the curve at the optimum: **steeper**, levels off **sooner**, at the **same** final level if the substrate is limiting.

Often asked as: explain each region of a rate–substrate curve; tangent for initial rate; conclusions from a pH × temperature table; immobilised v free enzyme data (6-mark).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Immobilised enzymes**, e.g. lactase trapped in **alginate** beads (WBI11 Jan 2026): can be **reused**; the product isn't contaminated with enzyme; active over a **wider range of pH and temperature** (higher optimum temperature), because "alginate holds the R groups in place / stabilises the active site", reducing vibration.

- **Rate equation** (given in the question): $V = V_{max} \times S \div (K + S)$, where K is the substrate concentration at $\frac{1}{2}V_{max}$ (WBI11 Jan 2020).
- Below the optimum, R groups vibrate a little and slightly distort the active site, but the extra collisions outweigh this, so the rate still rises (WBI11 Jun 2022).

COMMON MISTAKE

"At high temperature the enzyme moves too fast to bind." No: above the optimum the **active site changes shape** because bonds in the **tertiary structure** break, so the substrate is **no longer complementary**. Below the optimum, enzymes are **not** denatured; they just have less kinetic energy.

Explaining only one region of a curve: cover the rising part **and** the plateau. Say "molecules", not "particles".

Reading values off the curve when a **tangent** is needed; the tangent must touch the curve at the stated time. Give units and sensible decimal places.

"Immobilised enzymes are more efficient" or "the curve shifts right" score nothing: say what happens to activity and why.

Claiming an exact optimum from widely spaced readings. In a data table, draw conclusions rather than describing each row.

SPEC 2.9 Nucleotides, DNA and RNA

WHAT THIS SPEC POINT ASKS

(i) Know the structure of mononucleotides (deoxyribose or ribose + phosphate + a base: T, U, A, C or G) and of DNA and RNA as polynucleotides joined by condensation, forming phosphodiester bonds. (ii) Know how complementary base pairing and hydrogen bonding between two strands form the DNA double helix.

Nucleotides

- A **nucleotide** = a **pentose sugar** + a **phosphate group** + a **nitrogenous (organic) base**.
- The base is joined to **carbon 1** of the sugar, the phosphate to **carbon 5**.
- **Condensation** between the **phosphate** of one nucleotide and the **sugar** of the next (at carbon 3) forms a **phosphodiester bond**. A chain of these is a **polynucleotide**, with a **sugar–phosphate backbone**.

DNA vs RNA

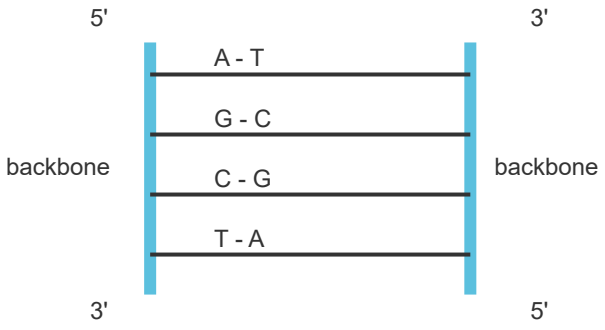
	DNA	RNA
Sugar	**deoxyribose**	**ribose**

	DNA	RNA
Bases	A, **T**, C, G	A, **U**, C, G
Strands	**two**, forming a **double helix**	**one**
Length	very long	much shorter
Job	stores genetic information stably	carries the code (mRNA), brings amino acids (tRNA), part of ribosomes (rRNA)

Bases are either **purines** (two rings: **A, G**) or **pyrimidines** (one ring: **C, T, U**).

Base pairing and the double helix

- The two DNA strands are held together by **hydrogen bonds** between **complementary** bases: **A–T** (2 hydrogen bonds) and **C–G** (3 hydrogen bonds). In RNA, A pairs with **U**.
- A purine always pairs with a pyrimidine, so the two strands stay a constant distance apart.
- The strands run in **opposite directions** (**antiparallel**) and twist into a **double helix**. The bases are on the inside, protected; the sugar–phosphate backbone is on the outside.
- Many hydrogen bonds together make DNA **stable**, yet each one is weak, so the strands can be **separated** for replication and transcription.



WORKED EXAMPLE

Worked example: base percentages

A DNA sample is 22% adenine. A pairs with T, so T = **22%**. A + T = 44%, so C + G = 56%, and C = G = **28%** each.

MARK-SCHEME WORDING

DNA only: **deoxyribose, thymine**. RNA only: **ribose, uracil**. Both: **phosphate, adenine, cytosine, guanine**. Write base names **in full**.

"**Phosphodiester** bonds formed by **condensation** (loss of water)"; complementary bases held by **hydrogen** bonds in a "**double helix**" ("helix" alone loses the mark).

Purine pairs with pyrimidine "so the strands are parallel", i.e. the helix has a **constant width**.

mRNA v tRNA: **both** single-stranded, ribose, A/U/C/G, phosphodiester bonds; mRNA is **linear**, of variable length, with **codons**; tRNA is **clover-leaf** shaped with **hydrogen bonds**, an **anticodon** and an **amino acid binding site**. "tRNA carries an amino acid" is ignored as a *structural* difference.

Bonds in double-stranded RNA: phosphodiester (sugar–phosphate), covalent (base–sugar), hydrogen (between bases): one mark each.

Often asked as: DNA / RNA Venn diagrams; draw a nucleotide, or two joined; compare mRNA and tRNA; base percentage calculations.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Nucleotides contain **C, H, O, N and P** (WBI11 Jun 2021).
- One turn of the helix is **3.4 nm** long and holds about **10 base pairs**, so adjacent base pairs are about **0.34 nm** apart (WBI11 Jan 2019).
- DNA mass and copies: after n divisions one molecule gives $\mathbf{2^n}$ copies; convert daltons to grams with the value given (1 Da = 1.67×10^{-24} g) (WBI11 Oct 2022).

COMMON MISTAKE

Phosphodiester bonds run **along** a strand; **hydrogen** bonds run **between** the strands. Don't mix them up when labelling diagrams, and don't join nucleotides in a strand by hydrogen bonds between their bases.

Drawing the base on the wrong carbon (it's C1), writing "P" for phosphate, or giving letters instead of base names.

Drawing a base pair when one nucleotide was asked for. In comparisons, write "**both**" for shared features.

SPEC 2.10 DNA replication and Meselson and Stahl

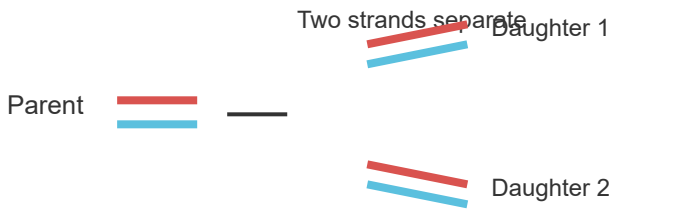
WHAT THIS SPEC POINT ASKS

(i) Understand DNA replication, including the role of DNA polymerase. (ii) Understand how Meselson and Stahl's experiment supported semi-conservative replication and refuted the competing theories.

Semi-conservative replication

1. The enzyme **DNA helicase** breaks the **hydrogen bonds** between bases and unwinds the double helix, separating the two strands.
2. **Each strand acts as a template.**
3. **Free DNA nucleotides** in the nucleus line up against the exposed bases by **complementary base pairing** (A–T, C–G), held by hydrogen bonds.
4. **DNA polymerase** joins adjacent nucleotides by **condensation reactions**, forming **phosphodiester bonds** in the new strand. It also **proofreads**, checking each new base pair and correcting mismatches.
5. Each new DNA molecule contains **one original strand and one new strand**: **semi-conservative** replication.

Replication happens before cell division, so each daughter cell gets an identical copy of the DNA.



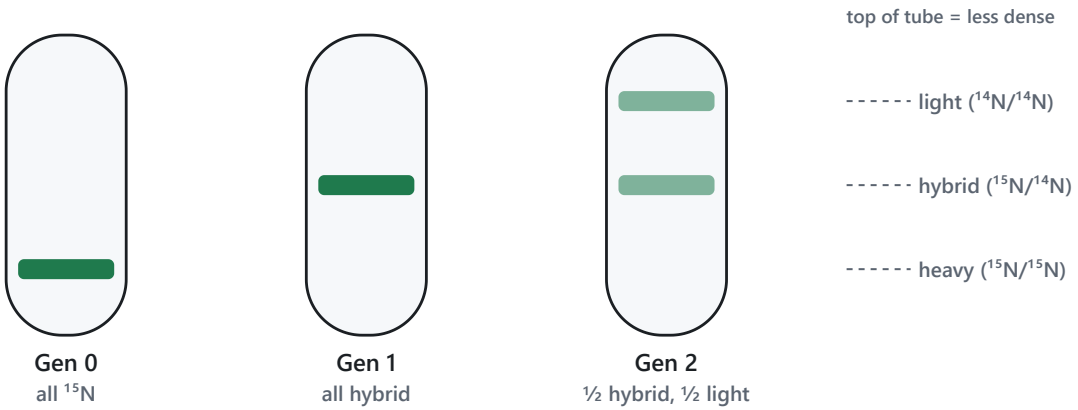
Red = original strand, blue = new strand (each daughter keeps c

Three competing theories

- **Conservative**: the original double helix stays intact; the copy is made entirely of new DNA.
- **Semi-conservative**: each new molecule is one old strand + one new strand.
- **Dispersive**: each new molecule is a patchwork of old and new sections along both strands.

Meselson and Stahl's experiment

1. Grew bacteria for many generations on a medium containing **heavy nitrogen (^{15}N)**, so all their DNA was "heavy" (nitrogen is in the bases).
2. Moved them to **light nitrogen (^{14}N)** and let them divide once, then twice.
3. Extracted the DNA each time and **centrifuged** it; heavier DNA settles **lower** in the tube.



Results: heavy band at the start; one intermediate (hybrid) band after one generation; one hybrid and one light band after two generations.

Generation in ^{14}N	Conservative predicts	Dispersive predicts	Semi-conservative predicts	Observed
1	two bands: heavy + light	one intermediate band	one intermediate band	**one intermediate band**
2	heavy + light (more light)	**one** band (between intermediate and light)	**intermediate + light** (equal amounts)	**intermediate + light**

- **Generation 1** rules out **conservative** (no heavy band).
- **Generation 2** rules out **dispersive** (two bands, not one).
- Only **semi-conservative** fits both.

Showing that replication is semi-conservative

1. Grow the bacteria in heavy nitrogen for several generations so that all their DNA is heavy.
2. Transfer them to light nitrogen. After one generation there is only an intermediate band. The conservative model is ruled out, because it would give a heavy and a light band.
3. After a second generation there is an intermediate band and a light band. The dispersive model is ruled out, because it would give only intermediate DNA.
4. Conclude that each new molecule contains one original strand and one newly made strand, which is the semi-conservative model.

EXAM TIP

For each model, state **what it predicts** and **why the result rules it out**. Explain the intermediate band: "each molecule has one ^{15}N strand and one ^{14}N strand, so it has an intermediate density."

MARK-SCHEME WORDING

"**Semi-conservative**": "**both** parent strands act as **templates**"; each new molecule has "**one parent strand and one new strand**". Explain both halves of the word.

DNA polymerase: lines up nucleotides along the template and "forms **phosphodiester bonds**". "Forms hydrogen bonds" is ignored. If asked for roles (plural), give more than one.

Enzymes: **helicase**, **DNA polymerase**, (DNA ligase). "RNA polymerase" is wrong here.

Meselson–Stahl as % of DNA (heavy / hybrid / light): start 100 / 0 / 0 → gen 1: 0 / 100 / 0 → gen 2: 0 / 50 / 50 → gen 3: 0 / 25 / 75. Conservative after one generation: heavy + light; dispersive: one intermediate band.

Replication gives "**genetically identical**" copies, so daughter cells make the same proteins. "Identical" alone isn't enough.

Often asked as: explain semi-conservative replication; roles of DNA polymerase; predict the bands for each model; complete a percentage table; read density-peak graphs.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Calculations of the **proportion of DNA** in each band after *n* generations: after 3 generations in ^{14}N, 2 of 8 molecules are hybrid and 6 are light → **25% hybrid, 75% light**. The number of hybrid molecules stays at 2 per original molecule.
- **DNA polymerase proofreading** keeps the mutation rate low (WBI11 Jan 2025).
- **DNA ligase** joins sections of new DNA together (WBI11 Jun 2019).
- **Replication bubbles** (WBI11 Jan 2022): replication starts at many points and runs in both directions. Copying 1.5×10^8 bases from one point at 50 bases s^{-1} would take $1.5 \times 10^8 \div 50 = 3 \times 10^6$ s ≈ **833 h**; with many origins it takes about 10 h.
- **Density graphs**: ^{15}N, hybrid and ^{14}N DNA give peaks at different densities, and peak height shows the **amount** of DNA (WBI11 Jan 2025).

COMMON MISTAKE

Mixing up "strand" and "molecule": each new **molecule** keeps one old **strand**.

Careless band drawings: put bands on the heavy, hybrid and light positions and show relative amounts by width. Fill every table cell (write 0, don't leave it blank).

Quick checks

QUICK CHECK

What type of bond joins two amino acids together in a polypeptide?

- Hydrogen bond
- Disulfide bridge
- Peptide bond
- Phosphodiester bond

Amino acids are linked by a peptide bond, formed by condensation between the carboxyl group of one and the amine group of the next. Hydrogen bonds and disulfide bridges hold the folded protein together rather than linking the chain, and phosphodiester bonds join nucleotides in DNA.

QUICK CHECK

Which pair of bases is joined by three hydrogen bonds in DNA?

- Adenine and thymine
- Adenine and guanine
- Thymine and cytosine
- Cytosine and guanine

Cytosine and guanine form three hydrogen bonds, while adenine and thymine form two. Adenine and guanine are both purines, so they cannot pair with each other, and thymine and cytosine are both pyrimidines.

QUICK CHECK

In the Meselson and Stahl experiment, what result after one generation in light nitrogen supports semi-conservative replication?

- Only a light band
- Equal heavy and light bands
- A single intermediate band
- Only a heavy band

Semi-conservative replication gives each new molecule one heavy and one light strand, which forms an intermediate band. A conservative model would have given half heavy and half light, and a light-only or heavy-only result would mean no replication had occurred.

QUICK CHECK

Which change would most directly denature a protein?

- Breaking the peptide bonds with water
- Adding a new amino acid to the end of the chain
- Heating it to a high temperature
- Increasing the length of the DNA gene

Heat disrupts the hydrogen, ionic and disulfide bonds that hold the folded shape, so the protein unfolds and loses function. Breaking peptide bonds is hydrolysis, not denaturation, and the other two options do not change the existing protein's shape.